

Serie 23

GRAM-SCHMIDT, ORTHOGONALITY

1. Let V be a finite dimensional euclidean vector space and $S \subset V$ an orthonormal set. Show that we can extend S to an orthonormal basis of V .
2. Let $A \in M_{n \times n}(\mathbb{R})$. Show:
 - (a) The matrix $A^T A$ is symmetric.
 - (b) The matrix $A^T A$ is positive definite if and only if A is invertible.
 - (c) We have $\text{rank}(A^T A) = \text{rank}(A)$.
 - (d) Assume that A is symmetric and that it admits $(\lambda_1, v_1), (\lambda_2, v_2)$ with $\lambda_1, \lambda_2 \neq 0$ as pairs of eigenvalue-eigenvector. Show that if $\lambda_1 \neq \lambda_2$ then $v_1 \perp v_2$.
3. Compute a decomposition $A = QR$ of the matrix

$$A = \begin{pmatrix} -1 & 1 & -1 \\ 2 & 0 & 1 \\ -2 & 1 & 0 \end{pmatrix}$$

into an orthogonal matrix Q and an upper triangular matrix R . Use this decomposition to solve the linear system $Ax = b$ for $b = (0, 3, -3)^T$.

4. Let U be the subspace of $\mathbb{R}^3$ with the standard inner product spanned by the two vectors $v_1 = (1, 1, 1)^T$ and $v_2 = (0, 2, 1)^T$.
 - (a) Determine an orthonormal basis of U and an orthonormal basis of $U^\perp$.
 - (b) Compute the representation matrices of the orthogonal projections $\mathbb{R}^3 \rightarrow U$ and $\mathbb{R}^3 \rightarrow U^\perp$ with respect to the standard basis of $\mathbb{R}^3$ and the bases from (a).

5. For each of the following vector spaces V endowed with the inner product $\langle \cdot, \cdot \rangle$, find $U^\perp$ for the given subset U :

(a) First consider

$$V = \left\{ (a_0, a_1, a_2, \dots) \mid \sum_{n=0}^{\infty} |a_n|^2 < \infty \right\}, \quad \langle (a_n)_{n=0}^{\infty}, (b_n)_{n=0}^{\infty} \rangle = \sum_{n=0}^{\infty} a_n \cdot b_n,$$

$$U = \{ (a_n)_{n=0}^{\infty} \in V \mid \exists N \geq 0 \text{ s.t. } \forall m \geq N : a_m = 0 \}$$

(b) Secondly, we set

$$V = C([0, 1]), \quad \langle f, g \rangle = \int_0^1 f(x) \cdot g(x) dx,$$

$$U = \left\{ f \in V \mid \int_0^{1/2} f(x) dx = 0 \right\}.$$

6. For a finite-dimensional Euclidean vector space $(V, \langle \cdot, \cdot \rangle)$ consider the isomorphism

$$\delta: V \rightarrow V^* := \text{Hom}_{\mathbb{R}}(V, \mathbb{R}), \quad v \mapsto \delta(v) := \langle v, \cdot \rangle.$$

- (a) Show that there exists exactly one inner product $\langle \cdot, \cdot \rangle^*$ on V^* such that δ is an isometry.
- (b) Let B be an ordered basis of V , and let B^* be the corresponding dual basis of V^* . Give the representation matrix of $\langle \cdot, \cdot \rangle^*$ with respect to B^* in terms of the representation matrix of $\langle \cdot, \cdot \rangle$ with respect to B .

Single Choice. In each exercise, exactly one answer is correct.

1. Consider the vector space $\mathbb{R}^2$ endowed with the standard scalar product $\langle \cdot, \cdot \rangle$ and the corresponding norm $\| \cdot \|$. For which vectors $v, w \in \mathbb{R}^2$ do we have $\|v + w\| = \|v\| + \|w\|$?
 - (a) $v = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, w = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$
 - (b) $v = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, w = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$
 - (c) $v = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, w = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$
 - (d) $v = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, w = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$
2. Let V be a euclidean vector space and let $v_1, v_2, v_3 \in V$. Which assertion does in general not hold?
 - (a) From $v_1 \perp v_2$ and $v_2 \perp v_3$ it follows that $v_1 \perp v_3$.
 - (b) From $v_1 \perp v_2$ and $v_1 \perp v_3$ it follows that $v_1 \perp (v_2 + v_3)$.
 - (c) From $v_1 \perp v_2$ it follows that $v_1 \perp -v_2$.
 - (d) From $v_1 \perp (v_2 + v_3)$ and $v_1 \perp v_2$ it follows that $v_1 \perp v_3$.
3. Let V be a euclidean vector space and let $S, T \subset V$ be two subsets. Which of the following properties is not equivalent to the other three?
 - (a) $S \subset T^\perp$
 - (b) $T \subset S^\perp$
 - (c) $S \perp T$
 - (d) $\langle S \rangle \cap \langle T \rangle = \{0\}$
4. Let S be a subset of a finite dimensional euclidean vector space V . Which asstertion does not hold in general?
 - (a) $(S^\perp)^\perp = \langle S \rangle$.
 - (b) S is the orthogonal complement of a subspace of V .
 - (c) $S^\perp$ is a subspace of V .
 - (d) $V = S^\perp \oplus (S^\perp)^\perp$.

Multiple Choice Questions

1. Which of the following matrices are hermitian?

(a) $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$

(b) $\frac{1}{i} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

(c) $\frac{1}{\sqrt{5}} \begin{pmatrix} i & -2 \\ 2 & i \end{pmatrix}$

(d) $\begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}$

2. Which of the following matrices are unitary?

(a) $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$

(b) $\frac{1}{i} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

(c) $\frac{1}{\sqrt{5}} \begin{pmatrix} i & -2 \\ 2 & i \end{pmatrix}$

(d) $\begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}$

3. Let U, V be unitary $n \times n$ matrices, and let $\lambda = e^{2\pi i\theta}$, $\theta \in \mathbb{R}$. In general, which of the following statements hold?

(a) $U + V$ is unitary.

(b) λU is unitary.

(c) U^{-1} is unitary.

(d) UV is unitary.